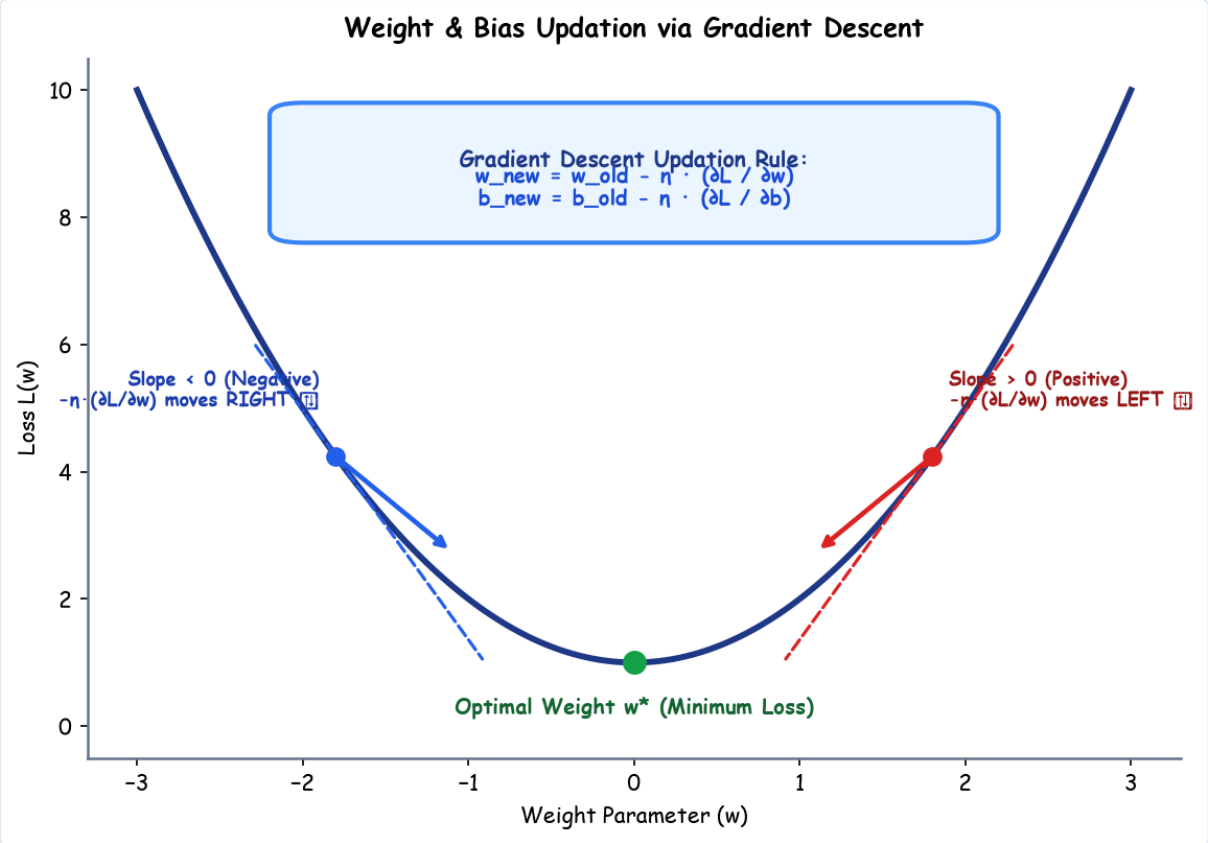
Submodule 37.04: Weight & Bias Updation

1. Introduction to Parameter Optimization

Once the loss function evaluates the discrepancy between predictions and ground truth, the network updates its internal parameter tensors (weights W and biases b) via **Gradient Descent**. Gradient Descent determines the magnitude and direction of adjustment necessary to move down the loss landscape towards the minimum.



2. The Core Parameter Update Equations

For any learnable weight parameter w and bias parameter b:

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot \frac{\partial \mathcal{L}}{\partial w}$$

$$b_{\text{new}} = b_{\text{old}} - \eta \cdot \frac{\partial \mathcal{L}}{\partial b}$$

Parameter Deconstruction

- $w_{\text{old}}, b_{\text{old}}$: Current numerical parameter values before optimization step.
- η (Eta): The **Learning Rate** hyperparameter (controls step size along the gradient vector).
- $\frac{\partial \mathcal{L}}{\partial w}, \frac{\partial \mathcal{L}}{\partial b}$: Partial derivatives representing the **Slope / Gradient** of the loss surface with respect to that parameter.
- The negative sign $(-)$ ensures movement occurs in the direction of **steepest descent**.

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3. Mathematical Mechanics: Why Does the Formula Always Converge?

Case 1: Point is on the Right of the Minimum (Positive Slope)

- Slope $\frac{\partial \mathcal{L}}{\partial w} > 0$ (Tangent angles upward).
- Formula calculation:

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot (+\text{value}) = w_{\text{old}} - \text{positive} < w_{\text{old}}$$

- **Result**: w decreases, moving **LEFT** $\leftarrow$ directly towards the minimum!

Case 2: Point is on the Left of the Minimum (Negative Slope)

- Slope $\frac{\partial \mathcal{L}}{\partial w} < 0$ (Tangent angles downward).
- Formula calculation:

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot (-\text{value}) = w_{\text{old}} + \text{positive} > w_{\text{old}}$$

- **Result**: w increases, moving **RIGHT** $\rightarrow$ directly towards the minimum!

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4. The Critical Role of the Learning Rate (η)

Learning Rate Setting	Dynamic Behavior on Loss Curve	Training Outcome
Too Large ($\eta \gg 0.1$)	Steps overshoot the global minimum, oscillating wildly and potentially diverging ($L \to \infty$).	Divergence / Exploding weights

Learning Rate Setting	Dynamic Behavior on Loss Curve	Training Outcome
Too Small ($\eta \ll 0.0001$**)**	Steps are microscopic; requires millions of iterations to converge.	Training stalls / Trapped in saddle points

| **Optimal (e.g.,** $0.001 - 0.01$**)** | Smooth, steady progression decelerating naturally as slope approaches zero. | Rapid, stable convergence |\n